

SIGNEX — SIGNAL EXTRACTION & ANALYSIS ENGINE

NTRO ID26147 CLASSIFIED INTELLIGENCE DOSSIER

CLASSIFICATION: SECRET // AIR-GAPPED INTERNAL // REF: f5b19ef166dbddae

INTERCEPT METADATA & CHAIN OF CUSTODY

Source Filename : sample_bpsk.iq
File SHA-256 Digest : f5b19ef166dbddae8b2878566e52571b0858b89aba1eb78684ad228b3ea007d
Payload File Size : 524,288 bytes (512.0 KB)
Analysis Timestamp : 2026-09-19T08:34:16Z
Operator ID : SIGINT-AIRGAP-OPERATOR
AMC Model Engine : amc_resnet18_v1.0 (Deep 1D ResNet-18 + Ensemble)

EXTRACTED RF SIGNAL PARAMETERS

Parameter Description	Estimated Value	Extraction Metric / Method
Sampling Frequency (F_s)	1.0000 MSps	Baseband Nyquist Rate
Center Frequency (F_c)	+0.001 kHz	Spectral Centroid Offset
Estimated Baud Rate	500.00 kBaud	Cyclostationary Wavelet Energy
3-dB Bandwidth	488.3 Hz	Half-Power Frequency Span
Occupied Bandwidth (99%)	849.61 kHz	Cumulative PSD 99% Energy
Estimated SNR	45.00 dB	Fourth-Order Cumulant (C_{42})
Peak-to-Average Power (PAPR)	2.68 dB	Temporal Envelope Crest Ratio
Carrier Frequency Offset (CFO)	+0.00 Hz	Non-Linear M-th Power Loop
Spectral Flatness (Entropy)	0.1379	Wiener Entropy Index (0.0 – 1.0)

PREDICTED MODULATION : BPSK

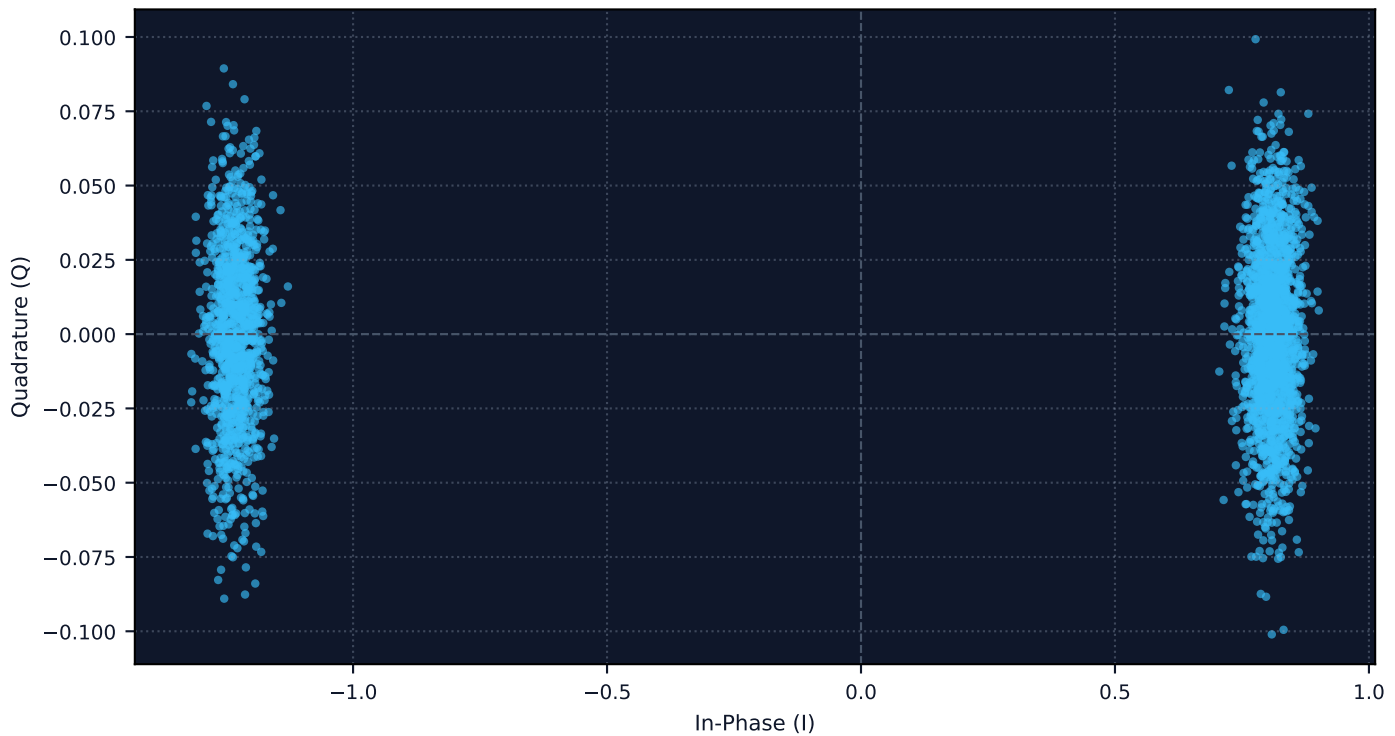
CLASSIFIER CONFIDENCE : 97.00% (ResNet-18 Softmax Ensemble)

INTELLIGENCE STATUS : CONFIRMED: HIGH CERTAINTY MODULATION CLASSIFICATION

Demodulation Scheme: BPSK | Recovered Bits: 8,192 | Estimated EVM: 10.00% | CRC Validated Frames: 0

SIGNAL DISSECTION & DEMODULATION INTELLIGENCE // [BPSK]

Demodulated Baseband Constellation Diagram — [BPSK]



SYNCHRONIZED TELEMETRY FRAMES & PROTOCOL DISSECTION

Frame #	Sync Index	Tx ID	Seq No.	Payload	CRC Status	FEC Status
#1	0	0x0000	0	1017 B	CRC MISMATCH	PARTIAL_RECOVER
#2	173	0x0000	1	995 B	CRC MISMATCH	PARTIAL_RECOVER
#3	340	0x0000	2	974 B	CRC MISMATCH	PARTIAL_RECOVER
#4	384	0x0000	3	969 B	CRC MISMATCH	PARTIAL_RECOVER
#5	429	0x0000	4	963 B	CRC MISMATCH	PARTIAL_RECOVER

DECODED BITSTREAM HEX / ASCII PREVIEW (FIRST 64 BYTES)

0000 48A82869E9C405E5E409882A87440647

[FRAME 1 PAYLOAD]